



INTERNATIONAL
STANDARD
UNIVERSITY
NEW DIMENSIONS IN EDUCATION



Claim Guard

Intelligent Invoice Verification & Anomaly Detection System

PROJECT TEAM

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THE PROBLEM

Manual invoice processing creates severe financial and operational risks.

Manual verification is slow, error-prone, and struggles across multiple data sources.



Duplicate Invoices

Paying the same invoice multiple times due to missing tracking.



Quantity Mismatches

Billed units exceeding the authorized Purchase Order limits.



Price Volatility

Unexpected spikes in unit prices compared to historical averages.



Identity Inconsistencies

Subtle vendor name or tax ID variations masking potential fraud.

The compounding cost of oversight.



Invoice Error



Financial Loss



Payment Disputes



Delayed Processing

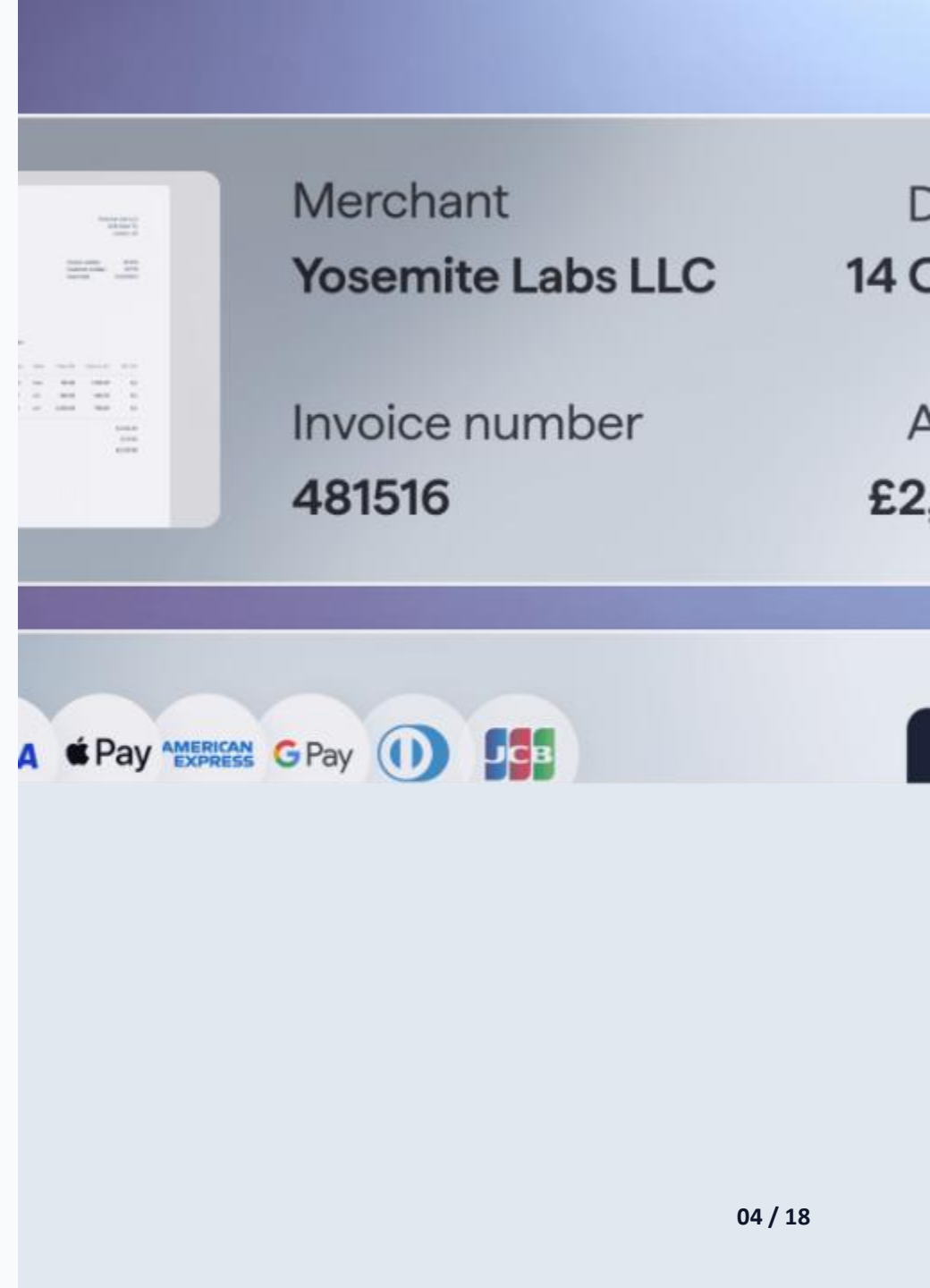
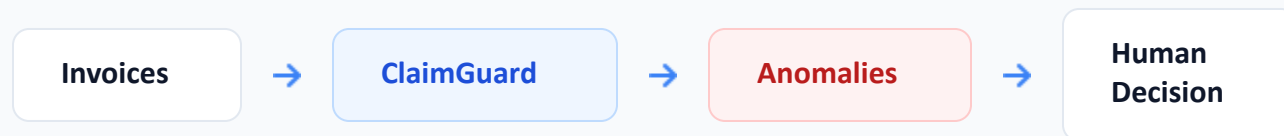


Operational Inefficiency

OUR SOLUTION

Meet ClaimGuard.

An intelligent invoice verification and anomaly detection system that helps reviewers identify suspicious or inconsistent invoices before payment.



CORE CAPABILITIES



Multi-Format Upload

Process PDF, Image, and Excel data.



Data Extraction

Capture vendor, amounts, taxes, and PO data.



Historical Comparison

Analyze past invoices for pricing baselines.



Duplicate Detection

Flag potential double-payment risks.



Quantity Verification

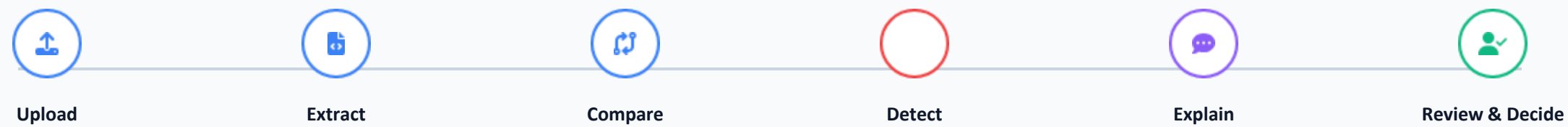
Cross-reference invoice against original PO.



Vendor Verification

Identify identity variations and mismatches.

HOW CLAIMGUARD WORKS



Real-world anomaly examples.

PRICE ANOMALY

Historical Average:₹480 – ₹500

Current Invoice:₹750

QUANTITY MISMATCH

Purchase Order:100 units

Invoice Quantity:150 units

IDENTITY VARIATION

Historical Record:"ABC Ltd"

Current Invoice:"ABC Limited Trading"

POSSIBLE DUPLICATE

Same Invoice Number

Same Total Amount

Same Vendor ID

EXPLAINABILITY

"Why was this flagged?"

ClaimGuard provides transparent reasoning for every alert. The human reviewer always makes the final business decision.

 **Invoice Flagged**

Risk Score: 78

DETECTION REASONS:

-  **Unit price is 48% above recent historical average.**
-  **Invoice quantity exceeds PO quantity by 50 units.**
-  **Vendor name differs from previous records.**

Request Verification

Reject Invoice

System Actors



Admin

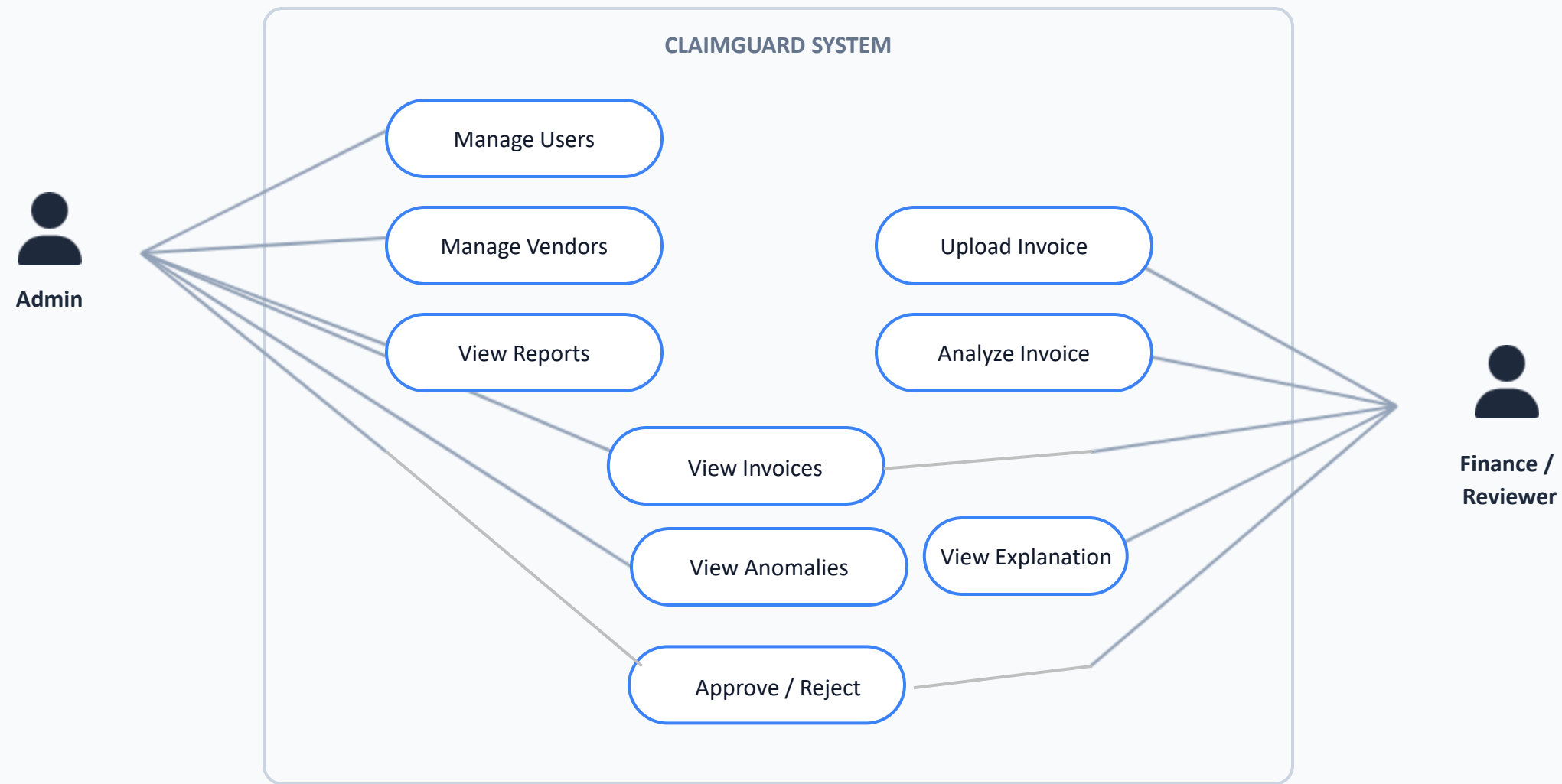
- ✓ Manage system users
- ✓ Manage vendor profiles
- ✓ View global invoices
- ✓ View anomaly metrics
- ✓ Generate reports



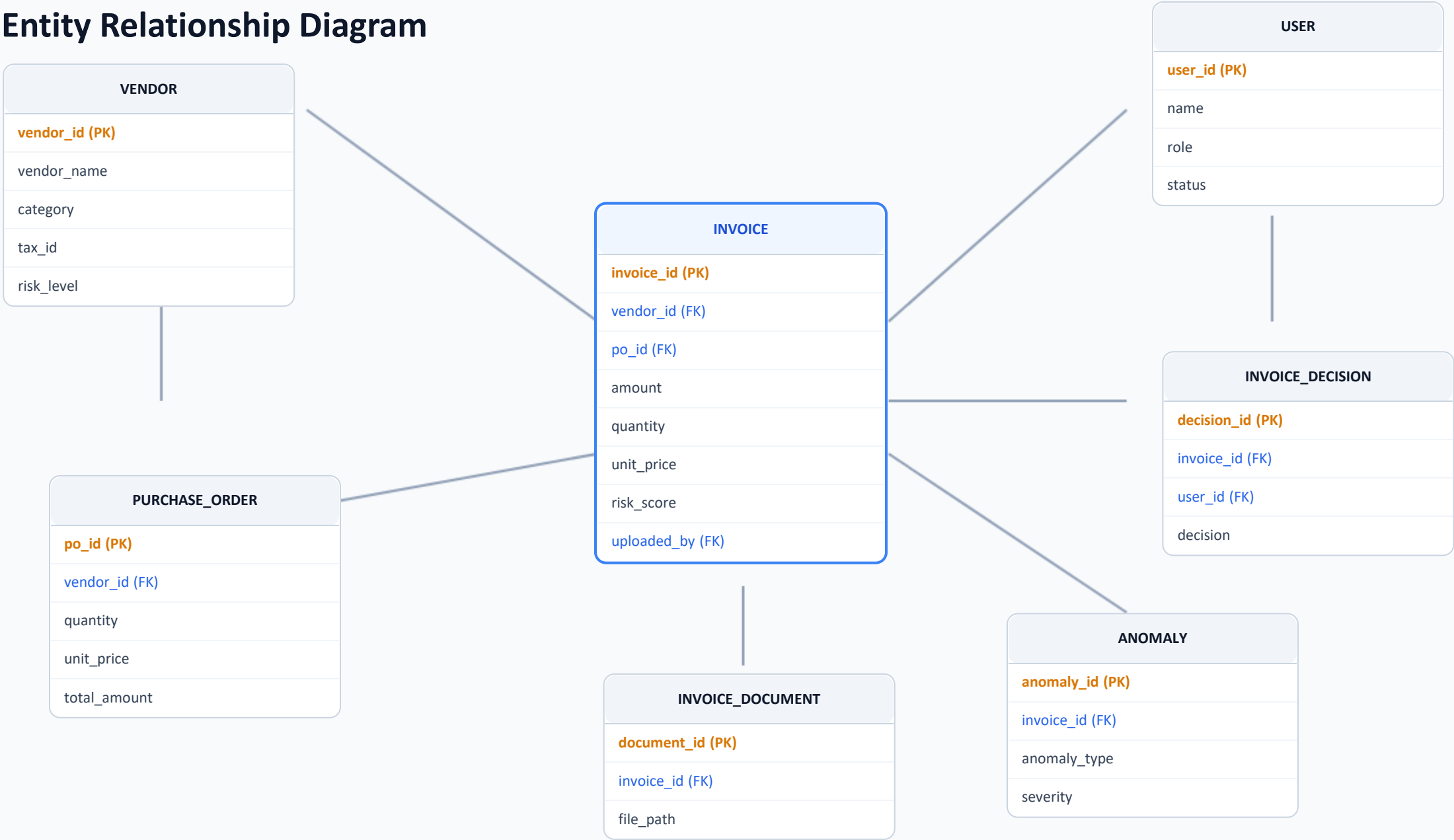
Finance / Reviewer

- ✓ Upload new invoices
- ✓ Analyze extracted data
- ✓ Review detected anomalies
- ✓ View explainability logs
- ✓ Approve, Reject, or Request Verification

Use Case Diagram

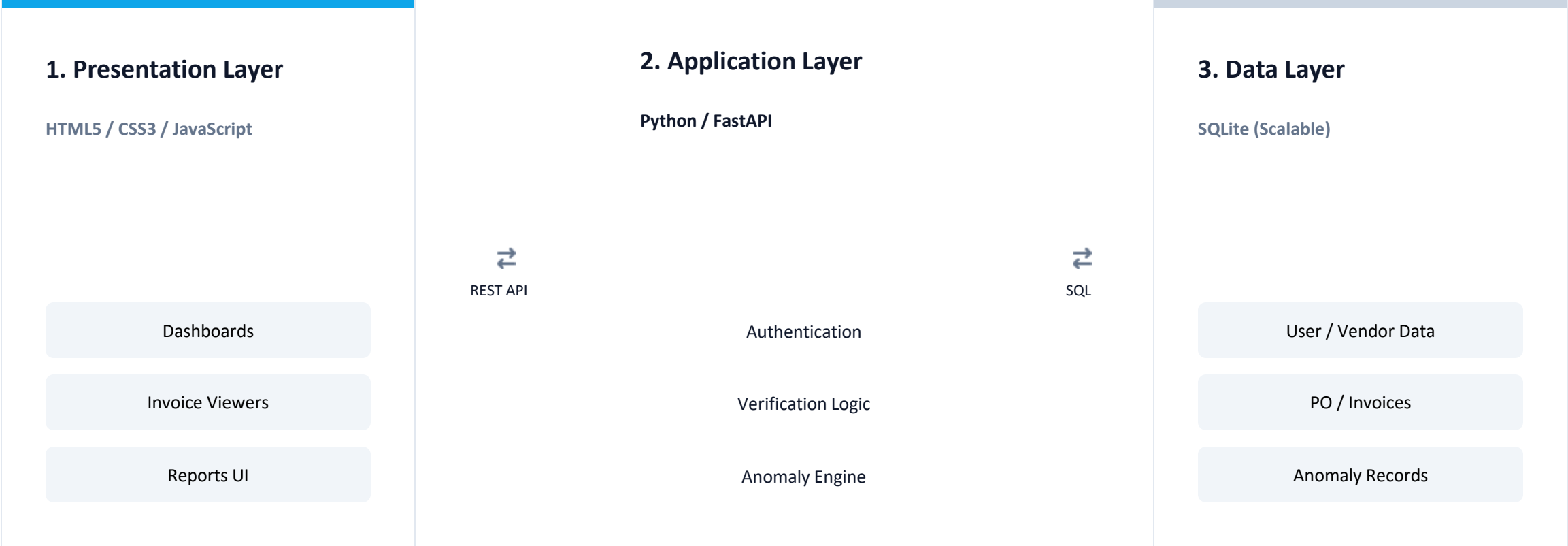


Entity Relationship Diagram



System Architecture

Clean Three-Layer Architecture.



Technology Stack



Frontend

HTML5
CSS3
JavaScript



Backend

Python
FastAPI
Uvicorn



Data & Tools

SQLite
Git / GitHub
VS Code

Limitations & Future Work

Current Limitations

- OCR accuracy constraints on poor-quality image uploads.
- Anomaly detection strictly relies on available historical data volume.
- Flags indicate statistical anomalies, not definitive proof of fraud.

Future Enhancements

- + Advanced Machine Learning models for context-aware extraction.
- + Adaptive anomaly thresholds based on seasonal vendor behavior.
- + Direct API integration with major ERP and accounting platforms.

Limitations & Future Scope

Current Prototype Limitations

- Does not guarantee fraud; requires human review.
- Detection quality depends strictly on data quality.
- Vendor-name similarity checks may yield false positives.
- Advanced Machine Learning requires larger datasets.

Future Roadmap

1. Current System

✓
2. OCR Invoice Extraction

🕒
3. Machine Learning Integration

🕒
4. ERP / Accounting Integration

🕒

Any Questions?

Let's talk about ClaimGuard.

THANK YOU

**Don't just process invoices.
Understand them before you pay.**
